

# Section: Resilience Case (Najm Application)

## 1 .The Failure Chain Analysis

To design for resilience, we must understand how a small bug can lead to a total service outage.

**Fault (The Defect):** A memory leak in the image-processing module used to upload accident photos.

**Error (Internal State):** The server’s RAM becomes exhausted, leading to an "Out of Memory" state for that specific microservice.

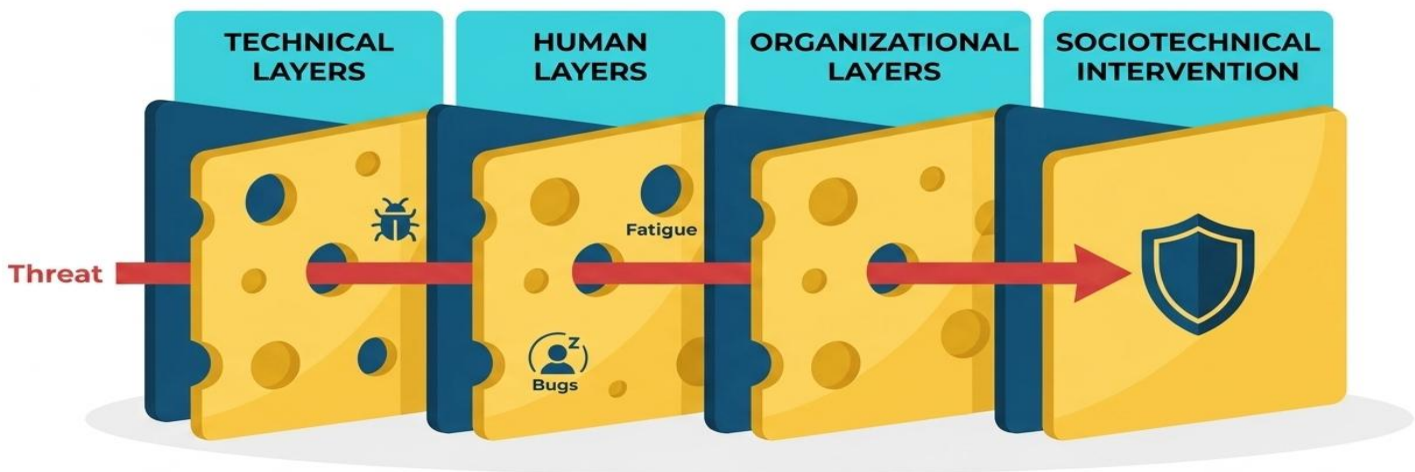
**Failure (Service Deviation):** The user is unable to submit the accident report, or the entire app crashes, preventing vital emergency assistance.

## 2. R&R (Recognition & Resistance) Table

| Strategy    | Technique for Najm App                                                                                                                                                                                  | Goal                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Recognition | <b>Heartbeat Monitoring &amp; Threshold Alerts:</b> Implementing an automated system that monitors CPU/RAM usage. If usage exceeds 90%, it triggers an instant alert.                                   | Detect the memory leak <b>before</b> the entire server crashes.                                            |
| Resistance  | <b>Graceful Degradation (Feature Isolation):</b> If the image-processing service fails, the app automatically disables photo uploads but keeps the <b>GPS Location and Text-based reporting</b> active. | Ensure the <b>critical service</b> (notifying a recovery agent) continues even if secondary features fail. |

## 3 .The Swiss Cheese Model (Defensive Layers)

### Swiss Cheese Model of System Failure



We visualize the Najm system as multiple layers of defense. A catastrophic failure only occurs if the "holes" (weaknesses) in all layers align.

**Layer 1: Technical (The Digital Barrier):** Automated load balancers and redundant cloud servers (AWS/Azure). Hole: A zero-day exploit or global cloud outage.

**Layer 2: Human (The Operational Barrier):** Highly trained dispatchers in the Najm Command Center. Hole: Human error or fatigue during high-volume periods (e.g., rainy days).

**Layer 3: Organizational (The Policy Barrier):** Strict Standard Operating Procedures (SOPs) for manual dispatching and offline data logging. Hole: Outdated emergency protocols.

Ensuring Non-Alignment: We use "Redundancy of Communication." If the Technical layer fails, the Organizational layer mandates switching to an offline radio-frequency (RF) communication system, ensuring the "holes" do not line up.

## 4 .The Sociotechnical Factor (Human-in-the-loop)

While Najm uses AI to assign investigators to accidents, human intervention is vital when automated systems reach their limits.

**Intervention Scenario:** In the event of a massive network blackout (System Failure), a Senior Dispatcher (Human-in-the-loop) takes over. They use a manual "Emergency Dashboard" to prioritize accidents based on severity (e.g., injuries vs. minor scratches), overriding the automated algorithm to ensure life-saving resources are deployed where they are needed most.